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Research Project Internship

Exploring Fragmentation Competition in Proteomics Data to Decode Chimeric Spectra

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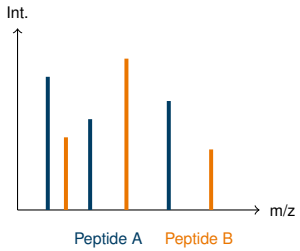
Chimeric Spectra in Proteomics

Traditional assumption: Each MS2 spectra originates from a single peptide.

Reality: Multiple peptides are often co-isolated and co-fragmented within the same isolation window, creating **chimeric spectra**.

Consequences:

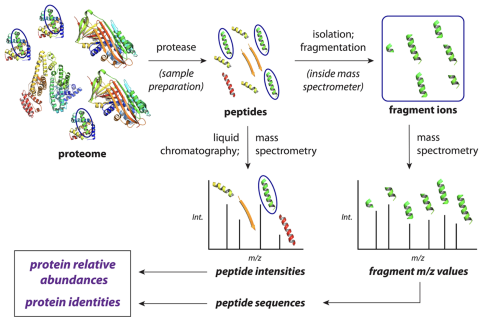
- Diluted peptide-specific signal
- Distorted scoring and missed identifications
- Errors propagate to protein quantification



Bottom-Up LC-MS/MS Proteomics

Workflow:

1. Proteins digested into peptides
2. Separation by liquid chromatography (LC)
3. Ionization (electrospray)
4. **MS1 scan:** Intact precursor ions (m/z, intensity)
5. **Isolation:** Select precursors in m/z window
6. **Fragmentation:** HCD $\rightarrow$ fragment ions
7. **MS2 scan:** Fragment ion spectra



Data-Dependent Acquisition (DDA):
Most intense precursors selected.

MSFragger-DDA+: Multi-Peptide Identification

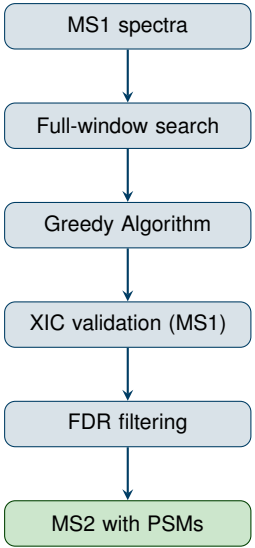
Traditional search engines:

- Assume one peptide per spectra
- Narrow precursor mass tolerance
- Miss co-fragmented peptides

MSFragger-DDA+ approach:

- **Full isolation-window search**
- **Hyperscore**
- **XIC extraction**: validates MS1 precursor evidence
- **FDR control**: PeptideProphet ≥ 0.99

$\Rightarrow$ Enables **multiple PSMs per spectra**



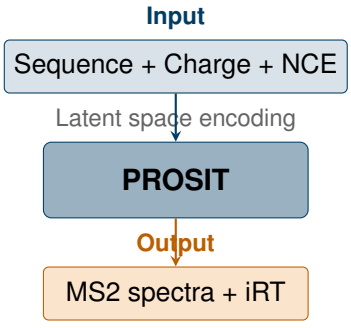
Prosit: Deep Learning spectra Prediction

What is Prosit?

- Deep learning model trained on ProteomeTools
- Predicts MS2 fragmentation patterns from peptide sequence

Workflow:

- **Input:** Sequence + Charge + NCE
- **Processing:** Latent space encoding
- **Output:** MS2 spectra + retention time (iRT)



Only b/y ions

Project Objectives

1. Assess whether MS1 intensity predicts MS2 fragmentation dominance
2. Evaluate a Sparse Regression Model to predict the composition of a Chimeric Spectra using predicted peptide from Prosit
3. Characterize deconvolution performance vs. spectral complexity

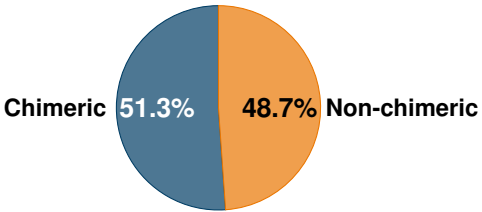
Dataset: Wide-Window Acquisition Benchmark

PXD037527

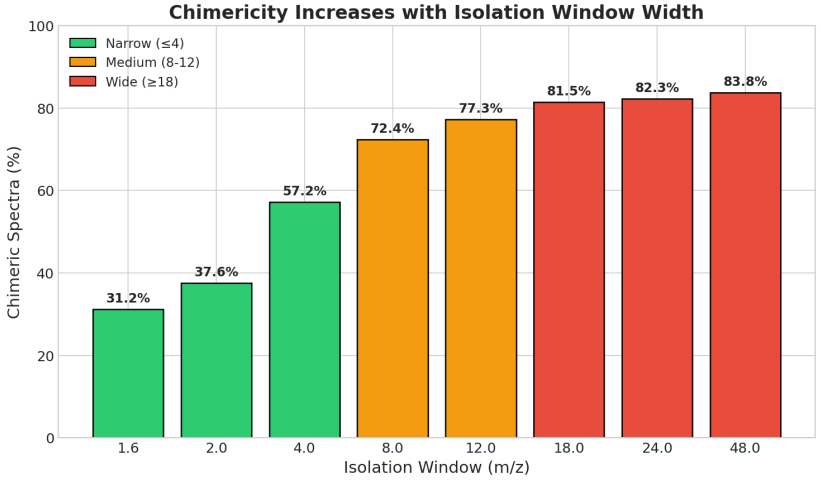
- 58 LC-MS/MS runs
- 8 Isolation windows: 1.6 – 48 m/z

MSFragger-DDA+ Output:

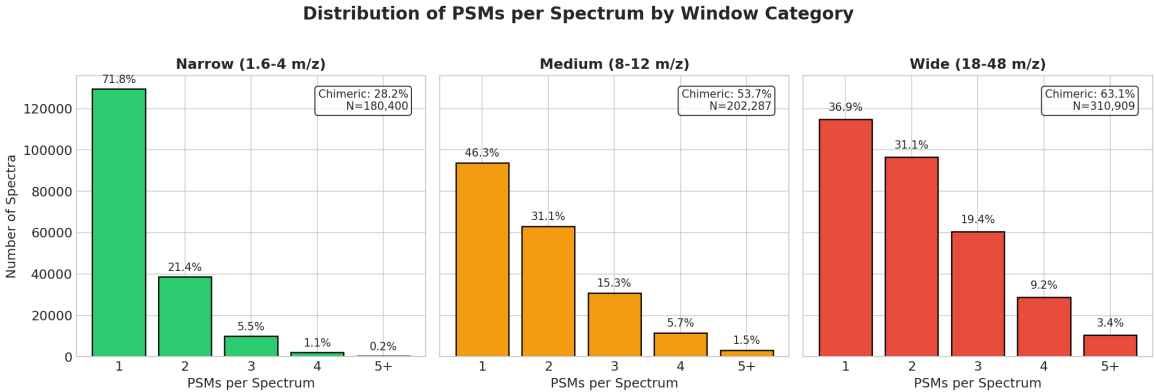
- 1,276,641 high-confidence PSMs
- 693,596 unique MS2 spectra



Chimericity Scales with Isolation Window Width



PSM Distribution Shifts with Window Width



Fragment Ion Annotation

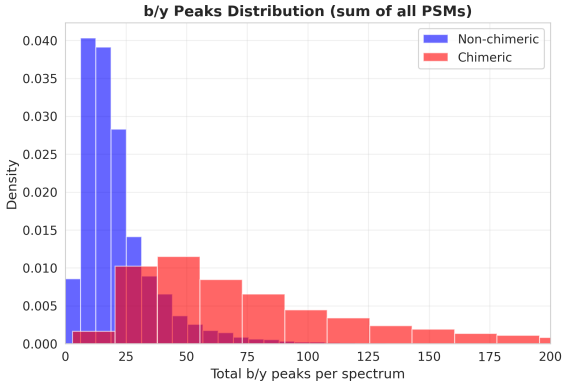
Annotation with spectrum_utils:

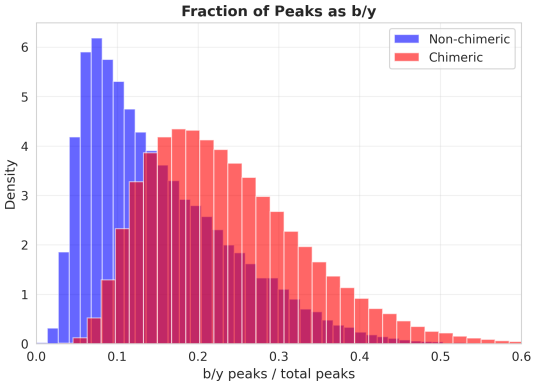
```
spec = Msmsspectrum(identifier,
    precursor_mz, precursor_charge,
    mz_array, intensity_array)
spec.annotate_proforma(
    peptide_sequence,
    fragment_tol_mass=20,
    fragment_tol_mode='ppm',
    ion_types='by',
    neutral_losses=False,
    max_ion_charge=precursor_charge)
```

Design choices:

- **b/y ions only**
- No neutral losses
- Charge $\leq$ precursor charge
- 20 ppm

Fragment Ion Annotation: b/y Peak Distribution





Fragment Ion Annotation: Interpretation

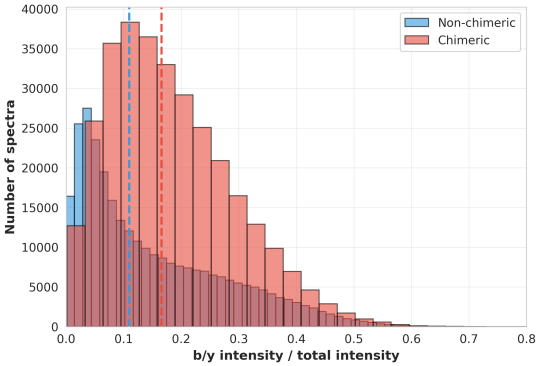
Key observations:

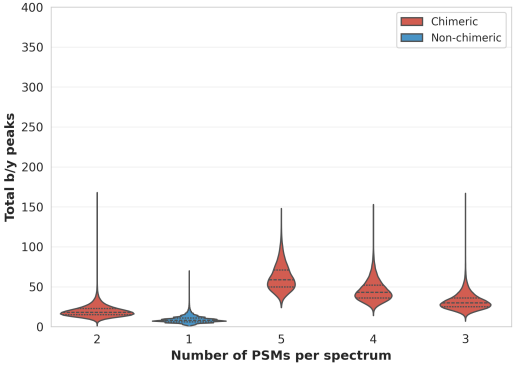
- Avg. **10.1** b/y ions per PSM
- Only **9.1%** of total spectral intensity
- Chimeric spectra show **3× more** b/y peaks

Interpretation:

- Majority of intensity remains *unassigned*
- High peak density not explained by peptide fragmentation alone

Fragment Signal Scales Linearly with Chimericity





b/y peaks increase **proportionally** with number of PSMs in the spectra

Defining Metrics

- **Hyperscore** : MSFragger metric (similarity score between observed and theoretical spectra, higher values indicate greater similarity)
- **FragShare** — fraction of annotated b/y intensity per peptide:

$$\mathrm{FragShare}_{s,p} = \frac{I_{s,p}^{\mathrm{frag}}}{\sum_{q \in \mathcal{P}_s} I_{s,q}^{\mathrm{frag}}}$$

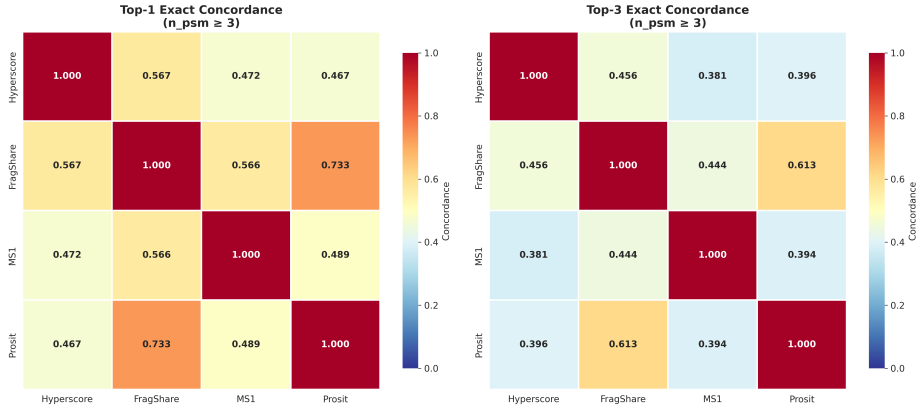
- **MS1 intensity** (IonQuant): ms1 abundance (area under the curve) of a PSM
- **Prosit cosine** — **similarity** between observed and predicted b/y intensities:

$$\cos(\mathbf{y}_{s,p}, \hat{\mathbf{y}}_{s,p}) = \frac{\mathbf{y}_{s,p}^{\top} \hat{\mathbf{y}}_{s,p}}{\|\mathbf{y}_{s,p}\|_2 \|\hat{\mathbf{y}}_{s,p}\|_2}$$

Question: Do these metrics agree on which peptide dominates?

Rank Concordance: Top1 and Top3 Metrics

Exact concordance



Rank Comparison findings: Top-1 vs Top-3

Key findings:

- FragShare–Prosit: highest agreement (**0.733**)
- MS1 vs all MS2 metrics: weak (**0.47–0.49**)
- **MS1 abundance does not predict fragmentation dominance**

Concordance Analysis: Interpretation

Top-1 concordance:

- FragShare & Prosit agree on dominant peptide (**0.733**)
- MS1 disagrees with all MS2 metrics ($\sim$**0.47**)

Top-3 concordance:

- Agreement drops to 0.38–0.61
- Methods agree on *set* of contributors, not *ordering*

Spectral Deconvolution: Problem Formulation

Goal: Decompose chimeric spectra into individual peptide contributions

Linear model:

$$\mathbf{y}_{\text{obs}} \approx \sum_{p \in \mathcal{P}_s} \beta_p \cdot \mathbf{X}_{\text{Prosit},p}$$

Design Matrix using Mass-channel representation:

- Channels defined only at Prosit-predicted m/z positions
- Mean: 57.6 channels per spectra

$$\mathbf{X}_s \in \mathbb{R}^{M \times |\mathcal{P}_s|}$$

LASSO Regression with Non-Negativity Constraint

Optimization problem:

$$\widehat{\beta}_s = \arg\min_{\beta \geq \mathbf{0}} \left\{ \frac{1}{2} \|\mathbf{y}_s - \mathbf{X}_s \beta\|_2^2 + \lambda \|\beta\|_1 \right\}$$

Components:

- $\mathbf{y}_s$: observed spectra (ℓ_1-normalized)
- $\mathbf{X}_s$: Prosit predictions (columns ℓ_1-normalized)
- $\beta \geq 0$: mixing coefficients (physical constraint)
- λ: regularization strength (induces sparsity)

Train/validation split: 70/30
Validation metric — reconstruction cosine:

$$\cos(\mathbf{y}_s, \hat{\mathbf{y}}_s) = \frac{\mathbf{y}_s^\top (\mathbf{X}_s \hat{\beta}_s)}{\|\mathbf{y}_s\|_2 \|\mathbf{X}_s \hat{\beta}_s\|_2}$$

Model Configurations

Three variants tested:

Model	Data	Design matrix
LASSO (2+ PSM)	All chimeric	Prosit only
LASSO (3+ PSM)	$\geq$3 PSMs	Prosit only
LASSO+MS1 (3+ PSM)	$\geq$3 PSMs	Prosit + MS1 prior

MS1 augmentation:

$$\mathbf{X}_s^{\text{aug}} = \begin{bmatrix} \mathbf{X}_s \\ \text{diag}(\mathbf{I}_{\text{norm}}^{\text{MS1}}) \end{bmatrix}, \quad \mathbf{y}_s^{\text{aug}} = \begin{bmatrix} \mathbf{y}_s \\ \mathbf{0} \end{bmatrix}$$

Adds soft constraint: β_p should approximate MS1 abundance

MS1 Augmentation: Example

Example: 2 peptides, 3 channels, MS1 = [0.6, 0.4]

Design matrix X^{aug}:

	Pep1	Pep2
ch1	0.3	0.1
ch2	0.2	0.4
ch3	0.1	0.3
ms1_1	0.6	0
ms1_2	0	0.4

Target vector y^{aug}:

	y
ch1	0.25
ch2	0.35
ch3	0.40
ms1_1	0
ms1_2	0

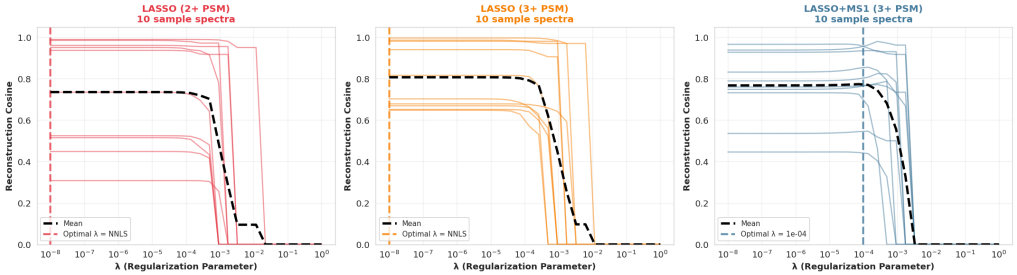
LASSO Training: Optimal λ Selection

Model	Training spectra	Optimal λ	Mean cosine $\pm$ SD	Active coeff.
LASSO (2+ PSM)	69,979	0 (NNLS)	0.79 ± 0.17	3.1
LASSO (3+ PSM)	46,767	0 (NNLS)	0.80 ± 0.15	3.6
LASSO+MS1 (3+ PSM)	35,830	10^{-4}	0.75 ± 0.15	3.2

Key observations:

- LASSO models select $\lambda = 0 \rightarrow$ **no sparsity needed**
- All PSMs are true contributors (already filtered by MSFragger)
- LASSO+MS1 requires $\lambda = 10^{-4}$ to mitigate MS1 bias

Regularization Paths: λ vs Reconstruction Quality



Interpretation:

- **LASSO (2+, 3+ PSM):** flat until $\lambda \sim 10^{-4}$, then collapse
- **LASSO+MS1:** suboptimal at $\lambda = 0$, peaks at 10^{-4}
- MS1 prior introduces bias $\rightarrow$ needs regularization to correct

Validation Performance: LASSO+MS1 Performs Worse

Model	Validation spectra	Cosine (mean)	Cosine (median)	$r(\beta,$ FragShare)
LASSO (2+ PSM)	29,991	0.795	0.840	0.661
LASSO (3+ PSM)	20,044	**0.803**	**0.843**	**0.703**
LASSO+MS1 (3+ PSM)	15,357	0.756	0.784	0.570

Key finding:

- LASSO (3+ PSM) achieves best performance
- Adding MS1 **degrades** reconstruction (0.803 $\rightarrow$ 0.756)
- MS1 abundance is not informative for deconvolution

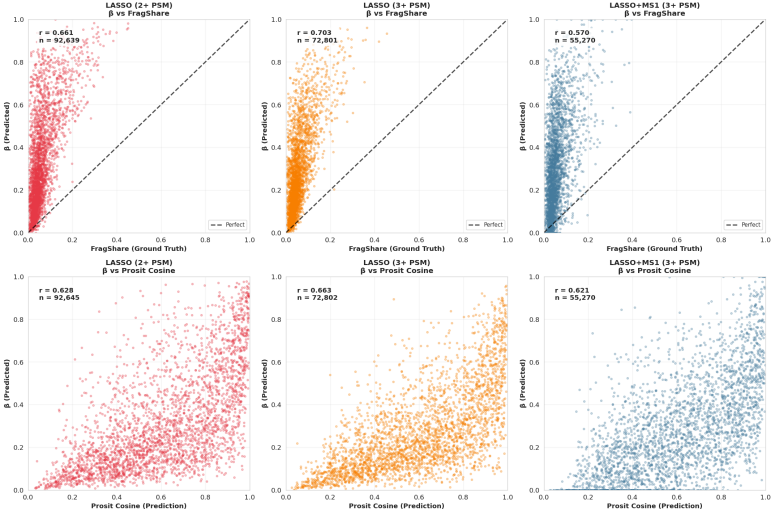
Validation Improves with Chimericity ?

Chimericity	LASSO (2+)	LASSO (3+)	LASSO+MS1
2 PSMs	0.778	–	–
3 PSMs	0.794	0.794	0.742
4 PSMs	0.806	0.806	0.767
5 PSMs	0.824	**0.827**	0.787
Trend	+6.0%	+6.3%	+6.1%

Counterintuitive finding:
More PSMs $\rightarrow$ better reconstruction

Why? More peptides = more constraints = better-conditioned system

Predicted β vs FragShare: Validation



Main Findings

1. **MS1 abundance does not predict MS2 dominance**
   - Concordance MS1 vs MS2 metrics: only 0.47
   - LASSO+MS1 performs worse than LASSO alone
   - Ionization efficiency $\neq$ fragmentation propensity
2. **Linear Deconvolution models work explain a resonably part**
   - Mean reconstruction cosine: 0.80
   - But: only 9% of intensity captured (b/y only)
3. **More chimericity $\rightarrow$ better reconstruction**
   - More peptides = better-conditioned problem ?

Why $\lambda = 0$?

LASSO selected no regularization — why?

Reason: PSM candidates already filtered by MSFragger-DDA+

- PeptideProphet probability ≥ 0.99
- All assigned PSMs are true contributors
- No false candidates to eliminate $\rightarrow$ no sparsity needed

Implication:

- LASSO reduces to NNLS (Non-Negative Least Squares)
- Sparsity would become necessary with more permissive filtering

Limitations

Data constraints:

- Only canonical b/y ions (9% of intensity)
- Prosit limitation to just b/y ions
- Chimericity

Model constraints:

- Linear assumption may miss competition effects
- High variance at single-spectra level

Future Directions

Test sparsity hypothesis:

- Use more permissive PSM filtering to get more Chimericity?
- Include more candidates? $\rightarrow$ LASSO should then sparsify?

Improve spectral coverage

Non-linear models

Different datasets

Acknowledgements

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